

1. Project Purpose

The point of this project is to make and implement a relational database for a small e-commerce store. The system has customers, payment cards, staff members, products, purchases and the individual products in each purchase. A console app that runs locally demonstrates the interaction with the db.

2. Project Scope

The system is made for a single store. Customers can look at products and make purchases. The staff can add products and update product info and restock inventory. This project mainly focuses on the backend database

3. User Roles

Role	Description	Main Capabilities
Customer	A shopper who purchases products.	Browse and search products, register payment cards, create purchases
Staff	An employee that does store operations.	Add products, edit product information, restock inventory

4. Data Requirements

- Customer data has unique customer id, first name, last name, email, phone, and account creation date.
- Credit card data has unique card id, owning customer, cardholder name, last four digits, expiration month, expiration year, and billing zip code. Full card numbers and security codes are not stored.
- Staff data: unique staff id first name, last name, unique email address, and job role.
- Product data has product id, product name, description, category, unit price, quantity in stock, and active status.
- Purchase data has purchase id, purchasing customer, processing staff member when applicable, purchase date, order status, and total amount.
- Purchase item data has purchase, product, quantity purchased, and the unit price at the time of purchase.

5. Requirements

Requirement
The system lets customers have unique email addresses.

Requirement
The system lets customers register multiple credit cards.
The system displays all active products available.
The system lets products be searched by name or category.
The system shall allow staff to add a new product.
The system lets staff update products.
The system can make purchases with more than one product.
The system rejects purchases when the item is out of stock.
The system reduces product inventory after a purchase.
The system calculates the total for a purchase.
The system display a customer purchase history.
The system supports SQL reports that have data from more than one table.

6. Nonfunctional requirements

- Data integrity has primary keys, foreign keys, unique constraints, check constraints, and transactions protect consistency.
- Security: The database doesn't store complete credit card numbers or security codes.
- Usability: The console menu has clear number choices and success or error messages.
- Reliability: Purchase creation and inventory reduction happen in one transaction.
- Maintainability: sql, application code, documentation, and sample queries are in clearly named files.

7. Business Rules

- A customer can have zero or multiple credit cards and each credit card belongs to only one customer.
- A customer can place no or many purchases and each purchase belongs to only one customer.
- A staff member may process zero or many purchases and a purchase may temporarily have no assigned staff member.
- A purchase must contain at least one purchase item.

- A product may appear in many purchase items and a purchase may contain many products.
- Product price and inventory quantity cannot be negative.
- Purchase item quantity must be greater than zero.
- The unit price recorded in PurchaseItem preserves the sale price even if the current Product price changes.
- A customer email and a staff email must be unique within their respective tables.

8. Primary Use Cases

Browse Products

Primary actor: Customer

1. Open the product menu.
2. System retrieves active products.
3. System displays product ID, name, category, price and stock.

Search Products

Primary actor: Customer

1. Enter a product name or category keyword.
2. System searches matching active products.
3. System displays matching products or a no results message.

Add Product

Primary actor: Staff

1. Enter product details.
2. System validates price and stock.
3. System inserts the product and confirms its new ID.

Create Purchase

Primary actor: Customer

1. Select a customer.
2. Enter one or more product IDs and quantities.
3. System verifies inventory.
4. System creates the purchase and purchase items, decreases inventory, calculates the total, and commits the transaction.

View Purchase History

Primary actor: Customer or Staff

1. Enter a customer ID.
2. System joins Customer, Purchase, PurchaseItem, and Product.
3. System displays previous orders and items.